# Control Barrier Functions

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## ABSTRACT

Exploring the Control Barrier Functions using gaussian kernels. Mainly debugging why they did not produce the expected outputs. Once the error was found, testing different length scales in order to dertime the optimal size for running on the indoor track.

Keywords: Keyword1, Keyword2, Keyword3

## INTRODUCTION

Control Barrier Functions(CBFs) are functions that can garentee the safety of the vehicle using lyaponov analysis. The CBFs used are based on the paper **?**. This paper demonstrates the use of Gaussian kernels to construct barrier functions in real time. The reason for the use of Gaussian kernels is to creates smooth differentiable barriers. The barriers are defined for all locations making feasability of the barriers more probable.

## METHODS AND MATERIALS

Guidelines can be included for standard research article sections, such as this one.

## SOME LaTeX EXAMPLES

Use section and subsection commands to organize your document. LaTeX handles all the formatting and numbering automatically. Use ref and label commands for cross-references.

### Figures and Tables

Use the table and tabular commands for basic tables — see Table 1, for example. You can upload a figure (JPEG, PNG or PDF) using the project menu. To include it in your document, use the includegraphics command as in the code for Figure 1 below.

| Item | Quantity |
|---|---:|
| Candles | 4 |
| Fork handles | ? |

**Table 1.** An example table.

### Mathematics

LaTeX is great at typesetting mathematics. Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{1}{n}\sum_{i}^{n} X_i$$

denote their mean. Then as $n$ approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

**Figure 1.** An example image of a frog.

### Lists

You can make lists with automatic numbering . . .

1. Like this,
2. and like this.

. . . or bullet points . . .

- Like this,
- and like this.

. . . or with words and descriptions . . .

**Word** Definition

**Concept** Explanation

**Idea** Text

## ACKNOWLEDGMENTS

Additional information can be given in the template, such as to not include funder information in the acknowledgments section.